

The vertex / MPM interface: particle-to-surface contact

prototype/ecm/test_03_mesh_contact.py · log/okuda_ECM/03_mesh_contact · 10 August 2026

1. The problem, and why the obvious scheme does not fit

The epithelium is a triangulated surface with no mass; the matrix and the basement membrane are MPM continua on a background grid. Every mechanism between them — adhesion, contact, load transmission — has to cross that boundary, and in this prototype it never has: the epithelium is a recorded surface that pushes and is never pushed back.

Outside biology the problem is standard, and it is called FEM–MPM coupling. The natural scheme, *grid-node* coupling (CFEMP, Lian et al. 2011), resolves contact by comparing the two bodies’ velocities at grid nodes they share, and it requires the mesh and the grid to be *comparable in size*. Ours are not. A cell is $0.73 \Delta x$ and the basement membrane $0.1 \Delta x$, so both bodies fall inside one cell, the grid hands them a single mass-weighted velocity, and they are welded: this is exactly what runs 142–151 measured ten times over. *Particle-to-surface* coupling (ICFEMP, Chen et al. 2015) removes the restriction by testing a particle against a mesh *face* rather than against a node, and that is what is implemented here.

2. The interface, in equations

For particle p under face f with outward normal $\hat{\mathbf{n}}$, let $\mathbf{w}$ be the barycentric weights of p ’s projection on f and $z_f = \sum_i w_i z_i$ the interpolated surface. The penetration, the normal penalty, and the regularised Coulomb friction are

$$d_p = [(\mathbf{x}_p - \mathbf{x}_f) \cdot \hat{\mathbf{n}}]_+, \quad \mathbf{f}_p^n = -k d_p \hat{\mathbf{n}}, \quad (1)$$

$$\Delta \mathbf{v}_t = (\mathbf{v}_p - \sum_i w_i \mathbf{v}_i)_{\perp \hat{\mathbf{n}}}, \quad \mathbf{f}_p^t = -\min\left(\mu k d_p, \mu k d_p \frac{|\Delta \mathbf{v}_t|}{\varepsilon}\right) \frac{\Delta \mathbf{v}_t}{|\Delta \mathbf{v}_t|}, \quad (2)$$

and the reaction is distributed to the face’s vertices by the same weights, which is what makes the exchange conservative by construction:

$$\mathbf{f}_i = -w_i (\mathbf{f}_p^n + \mathbf{f}_p^t), \quad \text{so} \quad \sum_p \mathbf{f}_p + \sum_i \mathbf{f}_i = \mathbf{0}. \quad (3)$$

ε is a slip velocity, not a fudge: it is the speed below which a contact is treated as stuck, so a resting contact does not chatter.

The one constraint that is not optional. The contact force reaches the particle as a body force, integrated at Δt , so it is stable only while $\Delta t \sqrt{k/m} < 1$, i.e.

$$k < k_{\max} = \frac{m}{\Delta t^2}, \quad m = \rho \Delta x^3 / \text{ppc} = 9.5 \times 10^{-7}, \quad k_{\max} = 95. \quad (4)$$

The first version of this rig used $k = 3 \times 10^4$, three hundred times past it. The stiffness is therefore written as a *fraction* of $k_{\max}$ — here 5%, so $k = 0.238$ — which is the same discipline the flat sheet needed after it NaN’d at $k = 6 \times 10^3$.

3. As a Plexus2 container

The contact is a *relation*, so it is an edge set and not a force field on a body:

object	kind	carries
<code>contact</code>	edge set, pre: <code>mpm_particle</code> , post: <code>vertex</code>	d_p , $\mathbf{w}$, bound flag
<code>mesh_contact</code>	Lateral on <code>contact</code>	returns a delta to <i>both</i> endpoints
<code>contact_rebuild</code>	Rewire on <code>contact</code>	which particle is under which face

Two placements matter. `mesh_contact` returns `{mpm_particle: f_p, vertex: f_i}` from one call, so Eq. (3) cannot be half-implemented — which is precisely how the spheroid’s adhesion lost its reaction. And it belongs *inside* the `substep_dt` block, recomputed at the substep positions: at frame level a stiff contact is integrated at Δt_{frame} and the ceiling in Eq. (4) falls by $(\Delta t_{\text{frame}} / \Delta t_{\text{sub}})^2 = 400$.

4. The rig, and what it measures

A flat patch of 17×17 vertices with four-neighbour springs, pressed at a prescribed speed into a block of 81,202 MPM particles; mesh cell $1.68 \Delta x$, no gravity. Three measurements, each of which kills the method if it fails.

	what it asks	result	reading
momentum	is the reaction really returned?	1.3×10^{-7} relative	machine precision in float32: <i>passes</i>
penetration	does the penalty bound it?	0.53 cells, 515 contacts	bounded, but half a cell is not small
slip	is this the grid's weld?	1.03×10^{-2} against 4.24×10^{-2}	friction changes it 4.1 $\times$: <i>it is not</i>

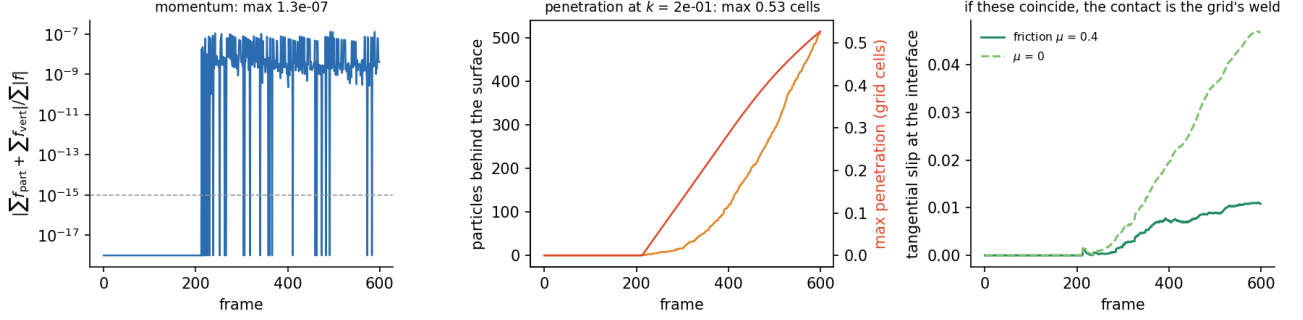


Figure 1: Left: the momentum residual, which is the test that a one-way coupling fails by construction. Middle: how many particles are behind the surface and how deep, in grid cells. Right: tangential slip with friction and without — if these two coincided, the “contact” would be MPM’s automatic weld wearing a friction coefficient.

Two failures the rig produced before it produced a result, both worth keeping. Under gravity the block *fell out from under the patch* — its floor is twenty cells below its underside, so it was in free fall and the two bodies never met (block top $0.5800 \rightarrow 0.5728$ while the patch went $0.5850 \rightarrow 0.5778$). And a patch of independent vertices is not a mesh: pressing its rim left the interior behind, so the lattice needed edge springs before a press could load anything.

5. What this does and does not license

It licenses the interface: a surface and a continuum can exchange force conservatively at a resolution where grid-node coupling cannot, with slip that is a material property rather than a numerical accident. It does not license the geometry — the patch is flat and axis-aligned, so face lookup is $O(1)$ and a real mesh needs a BVH — and it does not license the accuracy: half a grid cell of penetration is a lot, and the fix is either a stiffer penalty with a smaller substep or the barrier formulation (BFEMP, Li et al. 2022), which forbids penetration outright at the cost of an implicit solve. The next question is whether the same operator survives a *curved, moving* mesh, which is the spheroid.